

Lab07

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Laboratório 7

Item 1)

Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
# Criando a conexão e armazenando-a na variável conn
library(DBI)
library(RSQLite)
conn <- dbConnect(SQLite(), "uwmadison.sqlite3")
```

Item 2)

Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

Primeiro, quem são os professores:

```
SELECT DISTINCT instructors.name FROM subjects
INNER JOIN subject_memberships ON subjects.code = subject_memberships.subject_code
INNER JOIN sections ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.name
ORDER BY instructors.name
```

Table 1: Displaying records 1 - 10

name
ABIGAIL BENZINE
ADIN-CRISTIAN ANDREI
AKICHIKA OZEKI
ALAN HUANG
ALBERTO DEL PIA
ALBRECHT KLEMM
ALEXANDER COVINGTON
ALEXANDER FISH
ALEXANDER KISELEV
ALYSSA DIGILIO

Agora, quantos são:

```
SELECT COUNT(DISTINCT instructors.name) AS quantidade FROM subjects
INNER JOIN subject_memberships ON subjects.code = subject_memberships.subject_code
INNER JOIN sections ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
```

Table 2: 1 records

quantidade
377

Item 3)

O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

Quem é o professor mais difícil? Professor mais difícil seria aquele com o menor gpa O gpa de alguém = $(4.0A + 3.0AB + 3.0B + 2.5BC + 2.0C + 1.0D)/(A + AB + B + BC + C + D)$

```

SELECT DISTINCT instructors.name,
SUM(
    COALESCE(grade_distributions.a_count, 0) * 4.0 +
    COALESCE(grade_distributions.ab_count, 0) * 3.5 +
    COALESCE(grade_distributions.b_count, 0) * 3.0 +
    COALESCE(grade_distributions.bc_count, 0) * 2.5 +
    COALESCE(grade_distributions.c_count, 0) * 2.0 +
    COALESCE(grade_distributions.d_count, 0) * 1.0
)
/
NULLIF(
    SUM(
        COALESCE(grade_distributions.a_count, 0) +
        COALESCE(grade_distributions.ab_count, 0) +
        COALESCE(grade_distributions.b_count, 0) +
        COALESCE(grade_distributions.bc_count, 0) +
        COALESCE(grade_distributions.c_count, 0) +
        COALESCE(grade_distributions.d_count, 0)
    ),
    0
) AS GPA
FROM grade_distributions
INNER JOIN sections ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
INNER JOIN subject_memberships ON sections.course_offering_uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.name
HAVING GPA IS NOT NULL
ORDER BY GPA ASC
LIMIT 1

```

Table 3: 1 records

name	GPA
JONATHON PETERSON	2.820312

Quem é o professor mais fácil?

```

SELECT DISTINCT instructors.name,
SUM(
    COALESCE(grade_distributions.a_count, 0) * 4.0 +
    COALESCE(grade_distributions.ab_count, 0) * 3.5 +
    COALESCE(grade_distributions.b_count, 0) * 3.0 +
    COALESCE(grade_distributions.bc_count, 0) * 2.5 +
    COALESCE(grade_distributions.c_count, 0) * 2.0 +
    COALESCE(grade_distributions.d_count, 0) * 1.0
)
/
NULLIF(
    SUM(
        COALESCE(grade_distributions.a_count, 0) +
        COALESCE(grade_distributions.ab_count, 0) +
        COALESCE(grade_distributions.b_count, 0) +
        COALESCE(grade_distributions.bc_count, 0) +
        COALESCE(grade_distributions.c_count, 0) +
        COALESCE(grade_distributions.d_count, 0)
    ),
    0
) AS GPA
FROM grade_distributions
INNER JOIN sections ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
INNER JOIN subject_memberships ON sections.course_offering_uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.name
HAVING GPA IS NOT NULL
ORDER BY GPA DESC
LIMIT 1

```

Table 4: 1 records

name	GPA
YAJUAN SI	4

Qual é a disciplina mais difícil?

```

SELECT co.name AS disciplina,
       SUM(g.a_count*4 + g.ab_count*3.5 + g.b_count*3 + g.bc_count*2.5 + g.c_count*2 + g.d_count) AS gpa,
       SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_count) AS total
FROM grade_distributions g
JOIN course_offerings co ON g.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY co.name
HAVING GPA IS NOT NULL
ORDER BY GPA ASC
LIMIT 1

```

Table 5: 1 records

disciplina	GPA
Intro-Theory of Probability	2.886374

Qual é a disciplina mais fácil?

```

SELECT co.name AS disciplina,
       SUM(g.a_count*4 + g.ab_count*3.5 + g.b_count*3 + g.bc_count*2.5 + g.c_count*2 + g.d_count) AS gpa,
       SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_count) AS total
FROM grade_distributions g
JOIN course_offerings co ON g.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY co.name
ORDER BY GPA DESC
LIMIT 1

```

Table 6: 1 records

disciplina	GPA
Sample Survey Theory&Method	4

Item 4)

Desconecte do banco de dados.

```
dbDisconnect(conn)
```

Tempo

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-23 22:26:30 -03"
```